



NIFS FREE ONLINE COURSE · NIFS ES

Ergonomic Safety

Fit the task to the person.

Study guide & exam regulations

National Institute of Fire & Safety · nifsindia.net

Contents

01 Introduction

02 Training outcomes

03 Why it matters

04 Key benefits

05 Workplace hazards

06 Musculoskeletal disorders

07 Key principles

08 Assignment

09 Exam regulations & certificate

COURSE AT A GLANCE

Free No fees at any step	3 hours 2h study, 1h assessment	Online Study from anywhere	Certificate Emailed within 3 days of the exam
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1. **Register** name, mobile, email
2. **Study** the seven chapters
3. **Assignment** 10 marks
4. **Final exam** 20 marks, 20 questions, 10 minutes

Introduction: what is ergonomics?

Ergonomics is the field of study concerned with keeping people safe, comfortable and productive by accommodating human characteristics, capabilities and limitations in a product or process, at work and at home.

Rather than making the person adapt to fit the task, ergonomics fits the task to the person.

Also called human factors engineering, it is the scientific discipline of designing tools, tasks, environments and systems to fit the capabilities and limitations of people. It aims to optimise human well-being and system performance by reducing work-related musculoskeletal disorders (WMSDs), fatigue and discomfort, while maximising efficiency and productivity.

Training outcomes

Completing an ergonomic safety course reduces workplace injuries and boosts daily productivity by addressing occupational risks. After this course you will be able to:

- Understand the concept of ergonomics and its basic principles
- Identify ergonomic hazards in workstations
- Identify ergonomic risk factors and evaluate risk levels and severity
- Understand the importance of ergonomic safety
- Understand workplace design principles

Why ergonomic safety matters

Anyone on the job needs to know and understand:

- Ergonomics principles
- Ergonomic hazards in the workplace
- Ergonomic risks associated with the workplace
- Applying ergonomic principles in workplace design
- Primary and secondary factors of musculoskeletal disorders (MSDs)
- Material handling equipment
- Various environmental risks

Key benefits of the course

<i>Reduce risk and injuries</i> Prevent the awkward postures, repetitive strain and heavy lifting that cause musculoskeletal disorders (MSDs).	<i>Better posture and body mechanics</i> Practical techniques for safe lifting, sitting, standing and tool handling in offices and industrial settings.
<i>Early preparedness and hazard reporting</i> Recognise the early signs of physical strain and know when to report symptoms, before they become serious lost-time injuries.	<i>Lower direct and indirect costs</i> Preventing workplace pain reduces absenteeism, healthcare expenses and workers' compensation claims.

Ergonomic hazards at the workplace

Nine common hazards. If you can name them, you can spot them.

01 Poor workstation setup	02 Repetitive movements	03 Prolonged static postures
04 Forceful exertions	05 Vibration	06 Extreme temperatures
07 Inadequate lighting	08 Improper tools or equipment	09 Stress and workload

ERGONOMIC FACTORS

- Body posture
- Movement
- Environmental factors
- Information processing

ERGONOMIC-RELATED DISORDERS

- Musculoskeletal disorders (MSDs)
- Cumulative trauma disorders
- Repetitive motion injuries
- Back injuries
- Strains and sprains

Musculoskeletal disorders (MSDs)

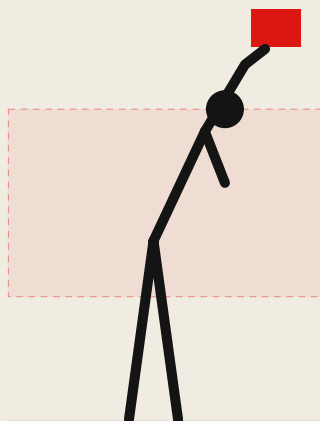
MSDs affect muscles, bones, joints, ligaments and tendons. Symptoms range from mild aches to severe issues, may worsen with activity or improve with rest, and often become persistent if ignored.

<i>Pain</i>	Aching, throbbing, sharp or dull pain, sometimes constant, sometimes intermittent.
<i>Stiffness</i>	Especially after rest or on waking, limiting movement.
<i>Weakness</i>	Loss of strength or increased muscle fatigue.
<i>Swelling</i>	Inflammation around joints or tissues, sometimes with warmth or redness.
<i>Reduced range of motion</i>	Difficulty bending or moving joints fully.
<i>Joint noises</i>	Clicking, popping or grinding sounds (crepitus).
<i>Numbness / tingling</i>	"Pins and needles", burning, or body parts "falling asleep".
<i>Tenderness</i>	Pain when a specific muscle or joint is touched.
<i>Fatigue</i>	Feeling tired or lacking energy in the affected areas.

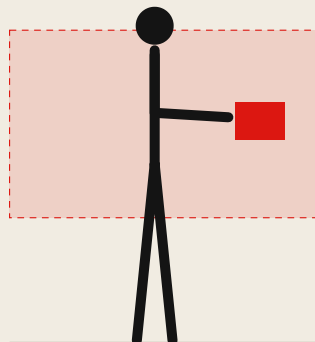
Notice it early. Report symptoms before they turn into a serious, lost-time injury.

Key ergonomic principles

Design work to fit the user: neutral postures, less force and motion, everything in the comfort zone.



Awkward. The person goes to the task.



Neutral. The task comes to the person, inside the power zone.

01 <i>Maintain neutral postures</i> Work in natural, relaxed positions. Avoid awkward wrist, elbow, shoulder or back angles.	02 <i>Work in the power zone</i> Keep frequently used tools and items within easy reach, between mid-thigh and shoulder height, to minimise reaching.	03 <i>Reduce excessive force</i> Lighten loads, use larger muscles and mechanical assists, or ask for help to avoid straining.
04 <i>Minimise excessive motion</i> Optimise tasks to reduce repetitive movements, which can cause conditions like tendinitis.	05 <i>Minimise static load</i> Avoid holding one position for long. Use jigs or clamps to reduce manual positioning.	06 <i>Minimise contact stress</i> Avoid hard or sharp surfaces pressing on the body, which can restrict nerves and blood flow.
07 <i>Allow movement and stretching</i> Build in breaks, exercise and position changes to reduce fatigue and improve muscle balance.	08 <i>Provide adequate clearance</i> Leave enough space under desks and around workstations for proper leg and body positioning.	
09 <i>Keep a comfortable environment</i> Control lighting (avoid glare), temperature, noise and vibration.		

Assignment: 10 marks

Answer both questions in your own words after reading the material. Submit them on the course page.

1. Why is ergonomic safety important at any workplace?

5 marks

2. Explain the symptoms of musculoskeletal disorders (MSDs).

5 marks

STEP 04

Exam regulations

1. 20 multiple-choice questions, 1 mark each. There is no negative marking.
2. You have 10 minutes. The clock starts when you press Begin and cannot be paused.
3. When the time is over, the exam is over. Your saved answers are submitted automatically and unanswered questions score zero.
4. Every question must be answered before you can submit early.
5. Closing or refreshing the page does not stop or reset the clock.
6. Do not leave the exam window. Leaving it is recorded as a warning.
7. One attempt only. Answers cannot be changed after submission.
8. After the exam, your certificate is sent to your registered email within 3 days.

YOUR CERTIFICATE

After your exam is over, we send your certificate to your email within 3 days.

Use the same email you registered with. Questions? WhatsApp +91 83743 40999.